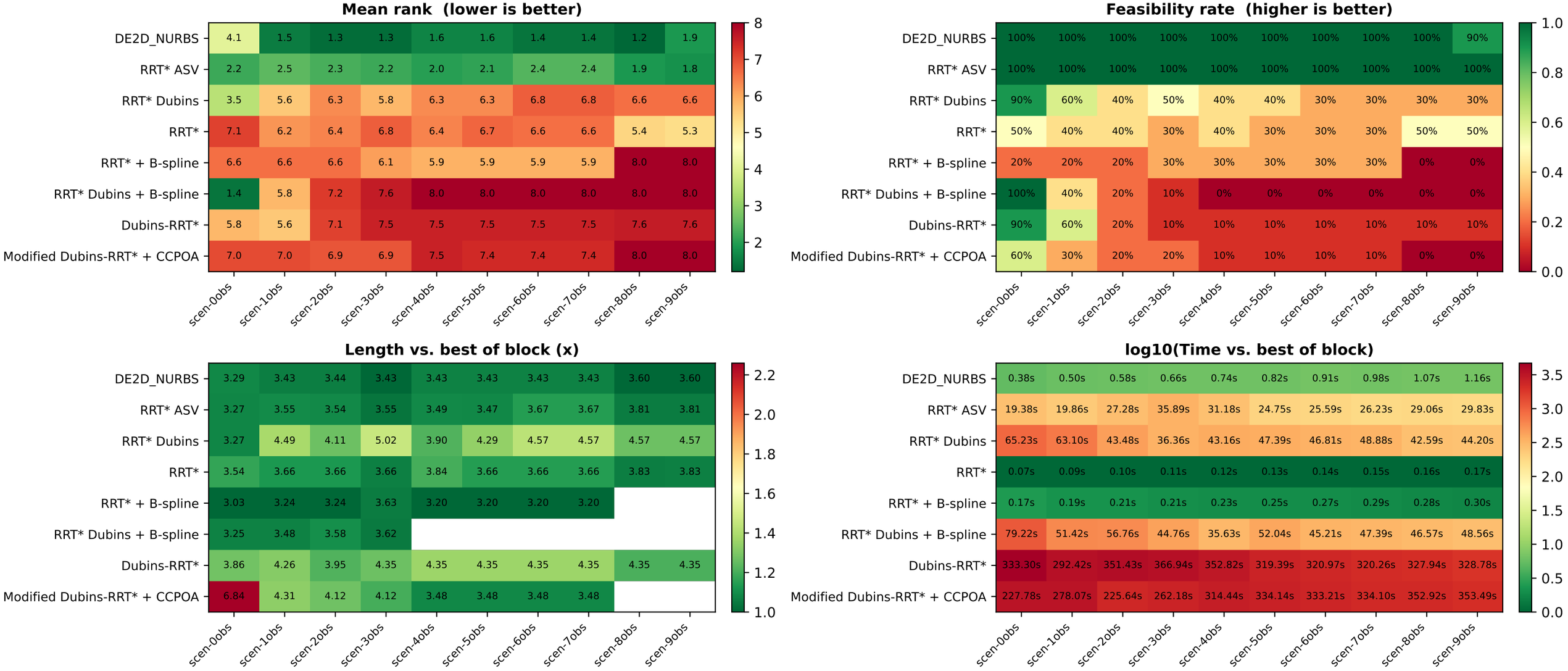

Mean rank (lower is better)
DE2D_NURBS 4.1 1.5 1.3 1.3 1.6 1.6 1.4 1.4 1.2 1.9
RRT* ASV 2.2 2.5 2.3 2.2 2.0 2.1 2.4 2.4 1.9 1.8
RRT* Dubins 3.5 5.6 6.3 5.8 6.3 6.3 6.8 6.8 6.6 6.6
RRT* 7.1 6.2 6.4 6.8 6.4 6.7 6.6 6.6 5.4 5.3
RRT* + B-spline 6.6 6.6 6.6 6.1 5.9 5.9 5.9 5.9 8.0 8.0
RRT* Dubins + B-spline 1.4 5.8 7.2 7.6 8.0 8.0 8.0 8.0 8.0 8.0
Dubins-RRT* 5.8 5.6 7.1 7.5 7.5 7.5 7.5 7.5 7.6 7.6
Modified Dubins-RRT* + CCPOA 7.0 7.0 6.9 6.9 7.5 7.4 7.4 7.4 8.0 8.0
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
8 7 6 5 4 3 2
Feasibility rate (higher is better)
DE2D_NURBS 100% 100% 100% 100% 100% 100% 100% 100% 100% 90%
RRT* ASV 100% 100% 100% 100% 100% 100% 100% 100% 100% 100%
RRT* Dubins 90% 60% 40% 50% 40% 40% 30% 30% 30% 30%
RRT* 50% 40% 40% 30% 40% 30% 30% 30% 50% 50%
RRT* + B-spline 20% 20% 20% 30% 30% 30% 30% 30% 0% 0%
RRT* Dubins + B-spline 100% 40% 20% 10% 0% 0% 0% 0% 0% 0%
Dubins-RRT* 90% 60% 20% 10% 10% 10% 10% 10% 10% 10%
Modified Dubins-RRT* + CCPOA 60% 30% 20% 20% 10% 10% 10% 10% 0% 0%
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
1.0 0.8 0.6 0.4 0.2 0.0
Length vs. best of block (x)
DE2D_NURBS 3.29 3.43 3.44 3.43 3.43 3.43 3.43 3.43 3.60 3.60
RRT* ASV 3.27 3.55 3.54 3.55 3.49 3.47 3.67 3.67 3.81 3.81
RRT* Dubins 3.27 4.49 4.11 5.02 3.90 4.29 4.57 4.57 4.57 4.57
RRT* 3.54 3.66 3.66 3.66 3.84 3.66 3.66 3.66 3.83 3.83
RRT* + B-spline 3.03 3.24 3.24 3.63 3.20 3.20 3.20 3.20
RRT* Dubins + B-spline 3.25 3.48 3.58 3.62
Dubins-RRT* 3.86 4.26 3.95 4.35 4.35 4.35 4.35 4.35 4.35 4.35
Modified Dubins-RRT* + CCPOA 6.84 4.31 4.12 4.12 3.48 3.48 3.48 3.48
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
2.2 2.0 1.8 1.6 1.4 1.2 1.0
log10(Time vs. best of block)
DE2D_NURBS 0.38s 0.50s 0.58s 0.66s 0.74s 0.82s 0.91s 0.98s 1.07s 1.16s
RRT* ASV 19.38s 19.86s 27.28s 35.89s 31.18s 24.75s 25.59s 26.23s 29.06s 29.83s
RRT* Dubins 65.23s 63.10s 43.48s 36.36s 43.16s 47.39s 46.81s 48.88s 42.59s 44.20s
RRT* 0.07s 0.09s 0.10s 0.11s 0.12s 0.13s 0.14s 0.15s 0.16s 0.17s
RRT* + B-spline 0.17s 0.19s 0.21s 0.21s 0.23s 0.25s 0.27s 0.29s 0.28s 0.30s
RRT* Dubins + B-spline 79.22s 51.42s 56.76s 44.76s 35.63s 52.04s 45.21s 47.39s 46.57s 48.56s
Dubins-RRT* 333.30s 292.42s 351.43s 366.94s 352.82s 319.39s 320.97s 320.26s 327.94s 328.78s
Modified Dubins-RRT* + CCPOA 227.78s 278.07s 225.64s 262.18s 314.44s 334.14s 333.21s 334.10s 352.92s 353.49s
scen-0obs scen-1obs scen-2obs scen-3obs scen-4obs scen-5obs scen-6obs scen-7obs scen-8obs scen-9obs
3.5 3.0 2.5 2.0 1.5 1.0 0.5 0.0